Solution to Assignment 1

1. (a) We only show that $\mathcal{F}_1$ is a σ-field, since the result can be obtained in the same way for $\mathcal{F}_2$. Set $A = \{1\}$. Then $\{2,3\}$ is A^c and it is obvious that $\mathcal{F}_1$ satisfies the first two conditions in the definition of a σ-field. As for $\cup_{i=1}^{\infty} A_i, A_i \in \mathcal{F}_1$, this union can only be $A, A^c, \emptyset, \Omega$, all of which belong to $\mathcal{F}_1$, verifying that $\mathcal{F}_1$ is a σ-field. Indeed, there are only the following 5 cases:
   i. $A_i = \Omega$ for some i. Then $\cup_{i=1}^{\infty} A_i = \Omega$;
   ii. $A_i = A$ and $A_j = A^c$ for some i, j. Then $\cup_{i=1}^{\infty} A_i = \Omega$;
   iii. (i) and (ii) don't hold and $A_i = A$ for some i. Then $\cup_{i=1}^{\infty} A_i = A$;
   iv. (i) and (ii) don't hold and $A_i = A^c$ for some i. Then $\cup_{i=1}^{\infty} A_i = A^c$;
   v. (i),(ii),(iii) and (iv) don't hold. In this case $A_i = \emptyset$ for all i. So $\cup_{i=1}^{\infty} A_i = \emptyset$.

   (b) $\{1,2\} \cap \{2,3\} = \{2\} \notin \mathcal{F}_1 \cup \mathcal{F}_2$, so $\mathcal{F}_1 \cup \mathcal{F}_2$ is not a σ-field.

2. (a) It is easily to find $\emptyset \cap Y = \emptyset$, so $\emptyset \in \mathcal{A} \mid Y$.

   (b) For any set $B \in \mathcal{A} \mid Y$, we can find a set $A \in \mathcal{A}$ such that $B = A \cap Y$. Notice that the complementary set of B with respect to Y is $A^c \cap Y$ and
   $$A^c \cap Y \in \mathcal{A} \mid Y.$$
   We know that $A^c \in \mathcal{A}$, so the complementary set of B over Y belongs to $\mathcal{A} \mid Y$.

   (c) For $B_i \in \mathcal{A} \mid Y$, we can also find $A_i \in \mathcal{A}$ such that $B_i = A_i \cap Y$. Notice that
   $$\bigcup_{i=1}^{\infty} B_i = \bigcup_{i=1}^{\infty} (A_i \cap Y) = \left(\bigcup_{i=1}^{\infty} A_i \right) \cap Y.$$
   Since $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$, we have $\bigcup_{i=1}^{\infty} B_i \in \mathcal{A} \mid Y$.

   Hence, $\mathcal{A} \mid Y$ is a $\sigma-$algebra over Y.

3. We set $A = \{1,2,3\}, B = \{3,4,5\}$. So $A^c = \{4,5\}$ and $B^c = \{1,2\}$ and $A \cap B = \{3\}$. So $\sigma(\mathcal{U}) = \{\{1,2,3\}, \{3,4,5\}, \{4,5\}, \{1,2\}, \{3\}, \{1,2,4,5\}, \Omega, \emptyset\}$.

4. For any interval $(a,b]$ with $a \leq b$, we have
   $$(a,b] = (a,+\infty) \cap (b,+\infty)^c \in \sigma(\mathcal{C}).$$
   So $\mathcal{B}(\mathbb{R}) \subset \sigma(\mathcal{C})$. On the other hand,
   $$(c,+\infty) = \bigcup_{n=1}^{\infty} (c, c+1).$$

By Proposition 1.1.11 in lecture notes, we know $(c, c+1) \in \mathcal{B}(\mathbb{R})$, so $(c, +\infty) \in \mathcal{B}(\mathbb{R})$ and $\sigma(\mathcal{C}) \subset \mathcal{B}(\mathbb{R})$. Thus $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R})$.